

Anemia Screening from Conjunctiva Images

Addressing Cross-Population Generalization, Device Invariance, and Data Leakage

Group-6

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Introduction

- ▶ Anemia affects over 1.9 billion people worldwide, especially young children and pregnant women.
- ▶ If left untreated, it causes severe complications like maternal mortality and permanent brain development delays in children.
- ▶ Doctors traditionally check for anemia by looking for paleness in the inner lower eyelid (conjunctiva).
- ▶ However, manual visual checking is subjective and varies greatly depending on room lighting and doctor experience.
- ▶ We aim to use smartphone computer vision to convert this visual check into an automated, objective screening tool.

Problem Statement

- ▶ Traditional blood tests (CBC) require painful needles, trained lab staff, and expensive blood analyzers.
- ▶ Rural clinics and remote screening camps often lack blood testing equipment, electricity, and lab reagents.
- ▶ Existing AI research papers claim 95% to 99.5% accuracy, but none of these systems are used in real hospitals.
- ▶ In reality, these models fail in real clinics due to patient data leakage and differences between phone cameras.
- ▶ Our goal is to build an honest, lightweight, and device-independent AI that runs directly on low-cost devices.

Comprehensive Survey of 40 Research Papers

We conducted an in-depth survey of 40 recent research papers across four core areas:

- ▶ Foundational clinical validation studies and public medical benchmark datasets.
- ▶ Deep learning architectures and Vision Transformers used for eye image analysis.
- ▶ Multimodal screening combining eye, fingernail, and palm images for hemoglobin estimation.
- ▶ Model quantization, edge computing deployment, and cross-population domain shifts.

Research Gap 1: Data Leakage (The Main Flaw)

- ▶ Many previous papers randomly split their dataset by image rather than by patient.
- ▶ This puts photos of the same person in both training and testing, so the AI simply memorizes the patient's face.
- ▶ On the identical 710 images (CP-AnemiC), the original creators got 84.8% accuracy, while later flawed papers claimed 99.5%.
- ▶ We will use strict patient-level grouping and automated tests to ensure zero data leakage and report honest results.

Research Gap 2: Cross-Population and Device Shift

- ▶ Almost all existing AI models are trained and tested in a single hospital with one camera type.
- ▶ Natural baseline eye color varies across ethnicities, genetics, and age groups, causing models to fail on new populations.
- ▶ Different smartphone brands (iPhone, Samsung, Xiaomi) process image colors differently, and room lighting changes tint.
- ▶ We will conduct the first zero-shot test across countries (Ghana to Italy/India) and use color constancy algorithms to fix lighting.

Research Gaps 3 & 4: Edge Deployment and Missing Inputs

- ▶ Many papers claim their models are suitable for mobile phones, but never test them on real low-power hardware.
- ▶ Standard compression methods (INT8 PTQ) cause severe drops in clinical sensitivity that remain unaddressed.
- ▶ Multi-site models (eye, nail, palm) crash completely if a patient only provides an eye photo (e.g. henna on hands).
- ▶ We will train models with modality dropout to handle missing photos, and use Quantization-Aware Training for edge hardware.

Project Objectives

- ▶ Measure and prove the exact accuracy difference between naive image splits and honest patient-level splits.
- ▶ Implement color constancy algorithms (Gray-World / Max-RGB) to remove camera-specific color tints.
- ▶ Evaluate zero-shot model performance across Ghanaian, Italian, and Indian patient datasets.
- ▶ Compress deep models to INT8 (~ 2.5 MB) using Quantization-Aware Training while keeping clinical sensitivity above 90%.
- ▶ Deploy and benchmark the final pipeline on a low-cost Raspberry Pi measuring real speed, memory, and energy.

① Automated Eye Segmentation:

- Crop the eye photo and isolate the reddish inner eyelid using color clustering and U-Net masks.
- Automatically detect and reject blurry, poorly lit, or out-of-focus photos before processing.

② Device-Invariant Color Correction:

- Apply color constancy filters to normalize lighting and remove camera-specific color casts.
- Extract key color indicators like the Erythema Index and CIELAB red-green balance.
- Augment training data with synthetic lighting and noise to simulate different phone cameras.

③ **Lightweight Dual-Head Backbone:**

- Train MobileNetV3-Small and EfficientNet-B0 to predict both anemia risk (yes/no) and exact hemoglobin level (g/dL).
- Train with modality dropout so the system works smoothly even when only one body part photo is provided.

④ **Model Compression & Edge Deployment:**

- Compress the model from 10 MB to ~ 2.5 MB using Quantization-Aware Training (QAT).
- Export to ONNX and TFLite runtimes for offline execution on a Raspberry Pi.
- Use Grad-CAM visual heatmaps to verify that the AI looks at blood vessels rather than eyelashes or skin.

▶ **CP-AnemiC Dataset (Ghana):**

- 710 eye images from children aged 6–59 months with lab-measured hemoglobin levels.
- Used as our primary dataset for model training and internal validation.

▶ **Eyes-Defy-Anemia Dataset (Italy & India):**

- 218 adult eye images from Italy and India with expert doctor segmentation masks.
- Used as our external test set to measure real-world performance on new populations.

▶ **Asare Multi-Site Dataset (Ghana):**

- Matched photos of eyes, fingernails, and palms from the same individuals for multi-input testing.

Tools Used & Edge System Architecture

Development Tools:

- ▶ Python 3.11, PyTorch, Torchvision, Albumentations, ONNX Runtime, and TFLite.
- ▶ Raspberry Pi 4/5 (low-cost 4,000 hardware target) and Jetson Nano.

System Workflow:

- ▶ Captures an eye photo using a standard smartphone or camera module.
- ▶ Runs the entire color correction, segmentation, and AI inference locally in under 50 ms.
- ▶ Works 100% offline without requiring internet access or cloud servers.
- ▶ Displays an immediate result: "Likely Anemic" or "Healthy" along with estimated hemoglobin (g/dL).

Project Timeline (2 Months / 8 Weeks)

- ▶ **Weeks 1–2 (Month 1):** Download datasets, set up workspace, and build strict patient-level splits with automated checks.
- ▶ **Weeks 3–4 (Month 1):** Build the eye segmentation and color constancy pipeline, and train baseline MobileNetV3 models.
- ▶ **Weeks 5–6 (Month 2):** Run cross-population benchmark tests (Ghana to India/Italy) and implement multi-site fusion.
- ▶ **Weeks 7–8 (Month 2):** Apply INT8 quantization, deploy on Raspberry Pi hardware, record latency and energy metrics, and write report.

Thank You!

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